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[22EC511]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

SATELLITE COMMUNICATION

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	Define Apogee and Perigee	L1	CO1	**02**
	b.	Define payload.	L1	CO3	**02**
	c.	Explain orbital perturbation.	L1	CO1	**02**
	d.	What is ionospheric effects?	L1	CO2	**02**
	e.	What is attitude control?	L1	CO3	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	With a neat diagram, explain Antenna Look Angles.	L2	CO1	**05**
		OR			
	b.	Define Geostationary Orbit. Mention the conditions required for an Orbit to be Geostationary	L2	CO1	**05**
3.	**a.**	Define Link Power Budget. Explain the Components of Link Power Budget	L2	CO2	**05**
		OR			
	b.	Explain the Transmission Losses	L2	CO2	**05**
4.	**a.**	Explain about the Transponder with a neat Block Diagram.	L2	CO3	**05**
		OR			
	b.	Explain Antenna Subsystem and the Earth Segment.	L2	CO3	**05**
5.	**a.**	Explain the various multiple access techniques	L2	CO4	**05**
		OR			
	b.	Explain the concept of GPS Navigation Message.	L2	CO5	**05**
6.	**a.**	Explain the concept of Navigation with Indian Constellation (NavIC).	L2	CO5	**05**
		OR			
	b.	Explain the concept of Interference between the satellite	L2	CO3	**05**
7.	**a.**	Mention and explain in brief the Applications of NavIC.	L2	CO5	**05**
		OR			
	b.	Explain the concept of Sun Transit Outage.	L2	CO1	**05**

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		PART-C	**6 X 10=60**		
8.	**a.**	Explain Earth Eclipse of Satellite with a neat diagram	L3	CO1	**10**
		OR			
	b.	Explain Kepler's Laws of Planetary Motion with necessary diagrams and mathematical expressions.	L3	CO1	**10**
9.	**a.**	Explain about Atmospheric Losses, Ionospheric Effects and Rain Attenuation with respect to Satellite Communication	L3	CO2	**10**
		OR			
	b.	Explain CNR and Combined CNR with Relevant Mathematical Equations.	L3	CO2	**10**
10.	**a.**	Explain the concept of Tracking, Telemetry and Command Subsystem.	L3	CO3	**10**
		OR			
	b.	With a neat Block Diagram, explain the concept of Receive Only Home TV System.	L3	CO3	**10**
11.	**a.**	Describe the TDMA , its features and efficiency	L3	CO4	**10**
		OR			
	b.	Explain how the bitrates are defined for digital TV	L3	CO4	**10**
12.	**a.**	Draw the general arrangement of position location with GPS and explain about GPS in detail	L3	CO5	**10**
		OR			
	b.	Explain GPS Receiver Operation.	L3	CO5	**10**
13.	**a.**	Define Antenna. With a neat diagram explain the concept of Horn Antenna and Parabolic Reflector.	L3	CO5	**10**
		OR			
	b.	Explain the Effects of Rain, Fade Margin and Rain Fade.	L3	CO2	**10**